`Problem 4` `Problems T4_PhaseDiagrams I.pdf #4` `Pb-Sn · hypoeutectic`

Phases and Microconstituents of a Pb-30 wt% Sn Alloy

`Original statement · Problems T4_PhaseDiagrams I.pdf`

1. *Using the Pb-Sn phase diagram for an alloy composition of Pb-30%Sn, indicate at the temperatures of $T_1 = 250$ ºC, $T_2 = 183 + \Delta T$ ºC, $T_3 = 183 - \Delta T$ ºC, $T_4 = 100$ ºC: a) The phases present, chemical composition of the phases and mass fraction. b) The micro constituents, chemical composition and mass fractions. Note the agreement that exists between the data obtained in (a) and (b).*

Official Solution: a) T_1: $C_L = 36\%$ **Sn,** $C_\alpha = 12\%$ **Sn,** $w_L = 0.75$; T_2: $C_L = 61.9\%$ **Sn,** $C_\alpha = 19.2\%$ **Sn,** $w_L = 0.25$; T_3: $C_\alpha = 19.2\%$ **Sn,** $C_\beta = 97.5\%$ **Sn,** $w_\alpha = 0.86$; T_4: $C_\alpha = 5\%$ **Sn,** $C_\beta = 100\%$ **Sn,** $w_\alpha = 0.74$; **b)** T_3: $C_{\alpha'} = 19.2\%$ **Sn,** $C_E = 61.9\%$ **Sn,** $w_E = 0.25$, $w_{\alpha'} = 0.75$

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Equilibrium cooling**, Pb-Sn diagram of the problem sheet (p. 1): eutectic at 61.9 wt% Sn and 183 °C; α of 19.2 and β of 97.5 wt% Sn at 183 °C.
2. **Readings of the figure:** at 250 °C the tie line meets the α solidus at 12 wt% Sn and the liquidus at 36 wt% Sn; at 100 °C the solvus lines give α of 5 and β of 100 wt% Sn (the same readings as the official solution).
3. $19.2 < C_0 = 30 < 61.9$: a **hypoeutectic** alloy, so the primary phase is α [Session 7 Slide 30]. This is the same kind of alloy as the hypoeutectic example of the slides.

Data

Temperature	Region	Phase compositions (wt% Sn)
$T_1 = 250$ °C	α + L	$C_\alpha = 12$, $C_L = 36$
$T_2 = 183 + \Delta T$	α + L	$C_\alpha = 19.2$, $C_L = 61.9$
$T_3 = 183 - \Delta T$	α + β	$C_\alpha = 19.2$, $C_\beta = 97.5$
$T_4 = 100$ °C	α + β	$C_\alpha = 5$, $C_\beta = 100$

PHASE 2 · FRAMES & STRATEGY

Frame: the Pb-Sn diagram (wt% Sn, °C); one horizontal tie line per temperature, so no change of basis is needed.

Method. Tie line and lever rule at each temperature [Session 7 Slides 15-16]; below 183 °C, the short tie line (α to eutectic liquid) gives the microconstituents and the long one (α to β) the phases [Session 7 Slide 32].

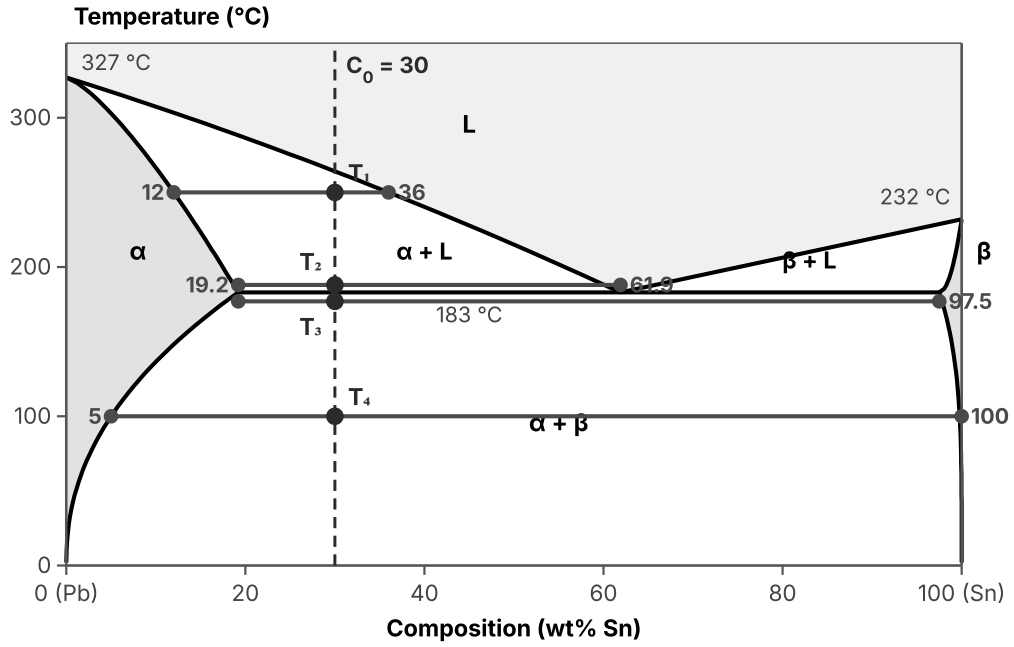


Figure 5. Pb-30 wt% Sn at T_1 = 250 °C (α of 12 and L of 36 wt% Sn, a reading of the figure), T_2 = 183 + ΔT (α + L), T_3 = 183 − ΔT (α + β) and T_4 = 100 °C. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 250$ °C. α + L; tie line from 12 to 36 (length 24):

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{30 - 12}{36 - 12} = \frac{18}{24} = 0.750 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{36 - 30}{24} = \frac{6}{24} = 0.250$$

$T_2 = 183 + \Delta T$**.** α + L; tie line from 19.2 to 61.9 (length 42.7):

$$W_L = \frac{30 - 19.2}{61.9 - 19.2} = \frac{10.8}{42.7} = 0.253 \qquad W_\alpha = \frac{61.9 - 30}{42.7} = \frac{31.9}{42.7} = 0.747$$

$T_3 = 183 - \Delta T$**.** α + β; tie line from 19.2 to 97.5 (length 78.3):

$$W_\alpha = \frac{97.5 - 30}{97.5 - 19.2} = \frac{67.5}{78.3} = 0.862 \qquad W_\beta = \frac{30 - 19.2}{78.3} = \frac{10.8}{78.3} = 0.138$$

$T_4 = 100$ **°C.** α + β; tie line from 5 to 100 (length 95):

$$W_\alpha = \frac{100 - 30}{100 - 5} = \frac{70}{95} = 0.737 \qquad W_\beta = \frac{30 - 5}{95} = \frac{25}{95} = 0.263$$

b) Microconstituents At T_1 and T_2 they coincide with the phases (α + liquid). At T_3: **primary α** (19.2 wt% Sn) and **eutectic** (61.9 wt% Sn overall):

$$W_{\alpha'} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{61.9 - 30}{61.9 - 19.2} = 0.747 = W_\alpha(T_2) \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{30 - 19.2}{42.7} = 0.253 = W_L(T_2)$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha'} = 0.862 - 0.747 = 0.115 \qquad W_{\beta,E} = W_\beta(T_3) = 0.138$$

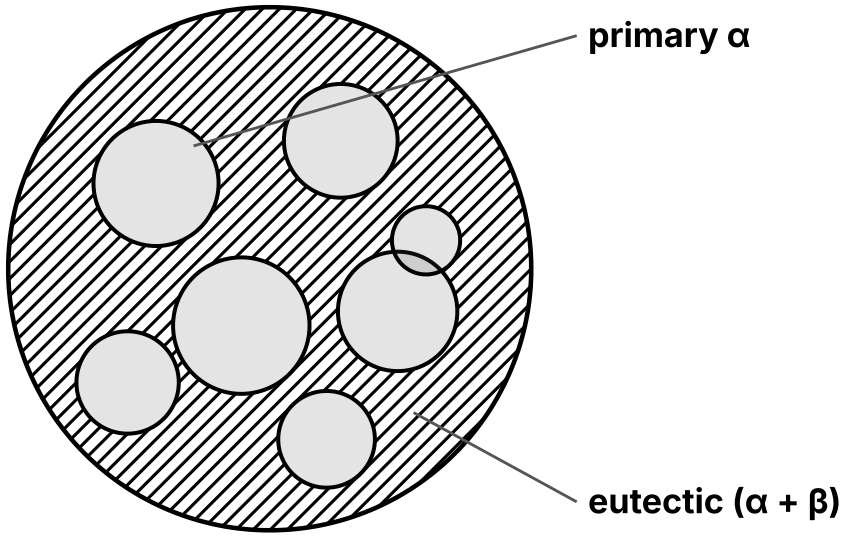


Figure 6. Pb-30 wt% Sn below the eutectic temperature: primary α (75 %) in the eutectic α + β (25 %).

PHASE 4 · CHECKS & INTERPRETATION

Agreement between (a) and (b)

- $W_{\alpha,E} + W_{\beta,E} = 0.115 + 0.138 = 0.253 = W_E$.
- Inside the eutectic, α is $0.115/0.253 = 0.455$, the eutectic product of the sheet diagram ($35.6/78.3 = 0.455$), the same as in Problem 3: the eutectic has the same make-up whatever the alloy.
- Mass balance at T_3: $0.862 \times 19.2 + 0.138 \times 97.5 = 16.55 + 13.45 = 30.0$ wt% Sn.

Interpretation The alloy is three quarters primary α in one quarter of eutectic, the structure of the hypoeutectic micrograph of Slide 30. From 250 °C to 183 °C the amount of α grows from 25 % to 75 % while the liquid gets richer in Sn, from 36 to 61.9 wt%.

Check against the official solution

The values agree with the key ($w_L = 0.75, 0.25$; $w_\alpha = 0.86, 0.74$; $w_E = 0.25$, $w_{\alpha'} = 0.75$). The α inside the eutectic is $0.862 - 0.747 = 0.115$; the worked page writes 0.11 because it subtracts the rounded values $0.86 - 0.75$.

ERRATUM IN THE WORKED SOLUTION

Solution problems Phase diagrams I.pdf, p. 10, gives for T_4 the numbers of Problem 3: $\chi_\alpha = (100 - 80)/(100 - 5) = 0.21$ and $\chi_{\beta,\text{total}} = 0.79$. For this alloy ($C_0 = 30$) the correct values are $W_\alpha = 70/95 = 0.74$ and $W_\beta = 0.26$, as the key of the statement says. The T_1 formula on p. 9 is also printed with the wrong labels ($C_o - C_\beta$ over $C_E - C_\alpha$), while its numbers, $(30 - 12)/(36 - 12) = 0.75$, are right.